

Uncertainty-Aware Fusion of Nighttime Light, Multispectral, SAR, and InSAR Imagery for Conflict Damage Assessment: A Bayesian Hierarchical Approach

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Abstract

Assessing infrastructure damage in conflict zones using satellite remote sensing is fundamentally challenged by the fact that no single sensor captures all dimensions of physical destruction. Nighttime light (VIIRS) detects functional collapse through power outages, multispectral imagery (Sentinel-2) captures spectral changes in vegetation and built structures, synthetic aperture radar (Sentinel-1) responds to structural scattering changes, and interferometric coherence (InSAR) measures surface stability. These four sensors measure fundamentally different physical quantities, and their damage assessments are often near-orthogonal in practice ($|\rho| < 0.11$ for all sensor pairs in our study). We propose a Bayesian hierarchical fusion framework that integrates these four heterogeneous data sources into a unified per-pixel damage probability with full uncertainty quantification via 95% highest density intervals. We further implement Dempster-Shafer evidence theory as a comparative fusion paradigm and systematically evaluate both against simple voting baselines. Applied to Mariupol, Ukraine—a city severely affected by the 2022 conflict—our framework produces a fused damage probability of 0.176 ± 0.032 (posterior mean $\pm$ posterior standard deviation), with a 95% HDI of $[0.059, 0.297]$. The near-zero cross-sensor correlations empirically validate the necessity of multi-source fusion, and the posterior uncertainty maps reveal where sensor disagreement is highest, providing actionable information for humanitarian assessment in data-scarce environments.

Keywords: Bayesian hierarchical model; multi-sensor fusion; damage assessment; remote sensing; Dempster-Shafer theory

1. Introduction

Satellite remote sensing has become an indispensable tool for monitoring armed conflicts and assessing their humanitarian impact. Since February 2022, the war in Ukraine has caused widespread infrastructure destruction, population displacement, and environmental degradation, creating an urgent need for reliable, scalable damage assessment methods. Remote sensing offers the unique advantage of accessing areas that are inaccessible to ground survey teams, providing synoptic, repeatable observations at multiple spatial and temporal scales.

However, a fundamental challenge persists: no single satellite sensor captures all dimensions of conflict damage. Nighttime light imagery (VIIRS/DNB) detects the *functional* collapse of urban areas through reduced electricity consumption (?), but cannot distinguish between power outages, population evacuation, and physical destruction. Multispectral imagery (Sentinel-2, Landsat) detects *spectral* changes in vegetation health and surface materials (?), but is confounded by seasonal phenology and cloud cover. Synthetic Aperture

Radar (Sentinel-1) amplitude responds to changes in surface roughness and dielectric properties, penetrating clouds but providing limited spectral information. Interferometric SAR coherence measures millimeter-scale surface stability, detecting structural collapse through phase decorrelation, but is sensitive to temporal baselines and atmospheric effects.

Each of these sensors measures a different physical manifestation of “damage,” and these manifestations are often decoupled in practice. A building reduced to rubble may show strong multispectral change (Sentinel-2) but negligible radar backscatter change (Sentinel-1), because rubble and intact buildings share similar C-band scattering properties. Conversely, a city-wide power outage produces dramatic nighttime light loss (VIIRS) with zero spectral or structural change. This sensor orthogonality means that single-sensor assessments are inherently incomplete, and that fusing multiple sensors is not merely beneficial but necessary for credible damage assessment.

In this paper, we propose a Bayesian hierarchical framework for fusing four heterogeneous satellite sensors into a unified, per-pixel damage probability with full uncertainty quantification. Our contributions are threefold:

1. We present the first four-sensor Bayesian fusion framework for conflict damage assessment, integrating VIIRS nighttime light, Sentinel-2 multispectral, Sentinel-1 SAR amplitude, and InSAR coherence data over a common study area.
2. We systematically compare three fusion paradigms—Bayesian hierarchical, Dempster-Shafer evidence theory, and simple voting—and analyze their behavior under varying degrees of sensor disagreement.
3. Through a case study of Mariupol, Ukraine, we demonstrate that the four sensors exhibit near-zero cross-correlation ($|\rho| < 0.11$), empirically validating the necessity of fusion and revealing the complementary nature of multi-sensor damage signals.

2. Related Work

2.1. Remote Sensing for Conflict Damage Assessment

Nighttime light (NTL) imagery from the VIIRS Day-Night Band has been widely used to detect conflict-induced urban decline. ? demonstrated that NTL changes correlate strongly with conflict intensity in Syria and Yemen, using monthly composites to track infrastructure collapse. ? extended this approach to quantify war-induced NTL loss across multiple conflict zones. However, NTL-based methods are limited to detecting functional rather than structural damage, and the 500 m spatial resolution of VIIRS prevents building-level assessment.

Multispectral change detection using Sentinel-2 and Landsat has been applied to conflict damage assessment through spectral indices such as NDVI, NBR, and NDBI (?). ? proposed a PCA-based multi-index fusion approach for building damage detection. The primary challenge is separating conflict-induced changes from seasonal phenology and inter-annual variability, which we address through same-season image pairing.

SAR-based damage detection exploits the sensitivity of radar backscatter to surface roughness and dielectric properties. ? used Sentinel-1 coherence change detection for building damage assessment, finding that coherence is more sensitive to structural damage than

amplitude alone. ? provide the foundational methodology for Sentinel-1 interferometric processing.

2.2. Bayesian Sensor Fusion

Bayesian hierarchical models provide a principled framework for combining information from multiple heterogeneous sources while propagating uncertainty through each level of the hierarchy (?). In remote sensing, Bayesian methods have been applied to land cover classification (?), change detection (?), and data assimilation. However, their application to multi-sensor conflict damage assessment remains unexplored.

2.3. Dempster-Shafer Evidence Theory

Dempster-Shafer (D-S) theory (?) generalizes Bayesian probability by allowing mass to be assigned to sets of hypotheses, not just singletons. This provides a natural framework for representing epistemic uncertainty—the “I don’t know” component that arises when sensors provide conflicting information. D-S has been applied to remote sensing classification (?) and multi-sensor fusion (?), but its comparative performance against Bayesian fusion in conflict damage contexts has not been studied.

3. Study Area and Data

3.1. Study Area

Our study area is Mariupol, a port city in southeastern Ukraine (47.1°N, 37.5°E) with a pre-war population of approximately 430,000. Mariupol experienced severe infrastructure damage during the 2022 conflict, with city officials reporting that over 90% of infrastructure was affected. The area of interest covers approximately 47 km (latitude) $\times$ 57 km (longitude), encompassing the urban core, surrounding suburbs, and agricultural hinterland.

3.2. Satellite Data

We use four satellite data sources spanning the period before and after the conflict (Table 1). Critically, for Sentinel-2 we use May-to-May (2021 vs. 2022) image pairs to eliminate seasonal phenology effects, a methodological choice validated by our finding that ΔNDVI changed from +0.141 (January–May, winter-to-spring) to -0.018 (May–May, inter-annual) when seasonal effects are controlled.

Table 1: Summary of satellite data sources used in this study.

Sensor	Product	Pre-war	Post-war	Resolution
VIIRS NTL	VNP46A2	2022-01-08	2022-05-08	500 m
Sentinel-2	MSI L2A	2021-05-23	2022-05-08	10–20 m
Sentinel-1	GRD IW VV	2021-06-01	2022-06-16	$\sim$ 10 m
InSAR Coh.	Interferogram	2021-12/2022-02	2022-02/2022-03	$\sim$ 20 m

4. Methodology

4.1. Overview

Our framework consists of three stages: (1) single-sensor damage probability estimation (Figure 1), where each sensor independently produces a per-pixel damage probability p_k and associated uncertainty σ_k ; (2) Bayesian hierarchical fusion, where the four sensor outputs are combined via a three-level Beta-Binomial conjugate model; and (3) comparative evaluation against Dempster-Shafer evidence theory and simple voting baselines.

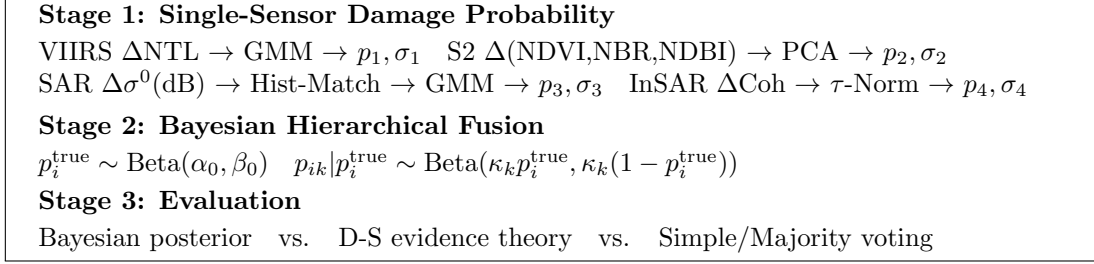


Figure 1: Three-stage fusion framework.

4.2. Single-Sensor Damage Probability Estimation

VIIRS Nighttime Light. We compute the difference in BRDF-corrected NTL radiance (ΔNTL) between pre-war and post-war images within the Mariupol AOI. A two-component Gaussian Mixture Model (GMM) is fitted to the ΔNTL distribution, where the component with the more negative mean is interpreted as the “damage” class. To prevent the GMM from being diluted by dark rural pixels, we fit the model exclusively on urban-core pixels (pre-war NTL $> 10 \text{ nW/cm}^2/\text{sr}$). The per-pixel damage probability p_1 is the posterior probability of belonging to the damage component. Bootstrap resampling ($B = 150$) over the urban-core pixels provides the uncertainty estimate σ_1 .

Sentinel-2 Multispectral. We compute three spectral indices—NDVI, NBR, and NDBI—and their differences (Δ) between May 2021 and May 2022. The three-dimensional Δ -vector is standardized and projected onto its first principal component (PC1, explaining 98.3% of variance). Crucially, we introduce a *directional damage score*: since PC1 loadings indicate that positive PC1 corresponds to vegetation increase (NDVI > 0 , NBR > 0 , NDBI < 0)—a non-damage direction—we compute $\text{score} = \max(0, -\text{PC1})$, which only activates for pixels changing in the damage direction (NDVI decrease, NBR decrease, NDBI increase). A GMM on this one-sided score then separates pixels with negligible change from those with significant damage-direction deviation.

Sentinel-1 SAR. We perform histogram matching between pre-war and post-war SAR VV-polarized backscatter to remove systematic radiometric biases caused by differing orbit geometries and seasonal moisture conditions. This step reduces the pre-post mean $\Delta\sigma^0$ from -1.61 dB (raw) to $+0.015 \text{ dB}$ (matched), confirming that the apparent darkening was a calibration artifact rather than a physical signal. A GMM is then applied to the matched dB difference to obtain p_3 and σ_3 .

InSAR Coherence. We normalize interferometric coherence using an exponential temporal decorrelation model: $\gamma_{\text{norm}} = \gamma \cdot \exp(\Delta t/\tau)$, with $\tau = 50$ days for C-band over mixed urban-agricultural terrain. This accounts for the different temporal baselines of the two interferograms (36 vs. 12 days). The normalized coherence difference is mapped to a damage probability via a sigmoid function.

4.3. Bayesian Hierarchical Fusion

We employ a three-level Bayesian hierarchical model with Beta-Binomial conjugacy. The model is defined as:

$$\begin{aligned}
 \text{Level 3 (Global Prior): } p_i^{\text{true}} &\sim \text{Beta}(\alpha_0, \beta_0), \quad \alpha_0 = \beta_0 = 1 \quad [\text{Uniform}(0,1)] & (1) \\
 \text{Level 2 (Sensor Likelihood): } p_{ik} \mid p_i^{\text{true}} &\sim \text{Beta}(\kappa_k \cdot p_i^{\text{true}}, \kappa_k \cdot (1 - p_i^{\text{true}})) & (2) \\
 \text{Level 1 (Posterior): } p_i^{\text{true}} \mid \mathbf{p}_i &\sim \text{Beta}\left(\alpha_0 + \sum_{k=1}^4 \kappa_k p_{ik}, \beta_0 + \sum_{k=1}^4 \kappa_k (1 - p_{ik})\right) & (3)
 \end{aligned}$$

where $\kappa_k = \min(1/(\hat{\sigma}_k^2 + 0.005), 30)$ is the sensor precision parameter, estimated from bootstrap resampling and regularized to prevent numerical instability. The Beta-Binomial conjugacy yields an analytic posterior, avoiding the need for MCMC sampling while retaining the full hierarchical structure. For each pixel, we compute the posterior mean $\mathbb{E}[p_i^{\text{true}} \mid \mathbf{p}_i]$, posterior standard deviation, and 95% Highest Density Interval (HDI).

4.4. Dempster-Shafer Evidence Theory

As a comparative paradigm, we implement Dempster-Shafer fusion over the frame of discernment $\Theta = \{D, U\}$ where D = “damaged” and U = “undamaged.” For each sensor k , the Basic Probability Assignment (BPA) is constructed as:

$$m_k(\{D\}) = p_k \cdot (1 - \sigma_k^{\text{norm}}), \quad m_k(\{U\}) = (1 - p_k) \cdot (1 - \sigma_k^{\text{norm}}), \quad m_k(\Theta) = \sigma_k^{\text{norm}} \quad (4)$$

where $\sigma_k^{\text{norm}} \in [0, 1]$ is the normalized uncertainty. The four BPAs are combined sequentially via Dempster’s rule of combination. The belief and plausibility of damage are reported, with the interval width $\text{Pl}(D) - \text{Bel}(D)$ quantifying epistemic uncertainty.

4.5. Baselines and Evaluation

We compare the Bayesian fusion against three baselines: (1) simple mean voting, (2) precision-weighted voting, and (3) median voting. For synthetic validation, we inject known damage patterns (three scenarios: VIIRS-dominant, S2-dominant, SAR-dominant, all-strong, and mixed) into background pixels and evaluate detection performance via ROC AUC.

5. Experiments

5.1. Single-Sensor Results

Figure 2 shows the damage probability maps from each sensor and the Bayesian fusion result. Table 2 summarizes the statistics.

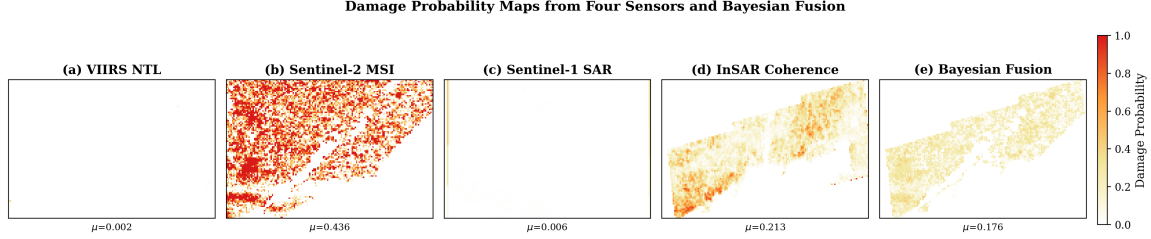


Figure 2: Damage probability maps: (a) VIIRS NTL ($\mu = 0.002$), (b) Sentinel-2 MSI ($\mu = 0.433$), (c) Sentinel-1 SAR ($\mu = 0.006$), (d) InSAR Coherence ($\mu = 0.215$), (e) Bayesian Fusion ($\mu = 0.176$).

Table 2: Single-sensor damage probability statistics.

Sensor	Mean p	Std p	Valid Pixels	Method
VIIRS NTL	0.002	0.020	14,015/15,000	GMM (urban-core fit)
Sentinel-2 MSI	0.433	0.030	11,051/15,000	PCA + directional GMM
Sentinel-1 SAR	0.006	0.031	15,000/15,000	Hist-match + GMM
InSAR Coh.	0.215	0.145	6,785/15,000	τ -normalized sigmoid

The VIIRS result reveals a critical spatial-scale effect: only 574 out of 76,047 pixels (0.8%) in the AOI had pre-war NTL > 10 nW/cm²/sr, corresponding to genuine urban land use. Within these urban-core pixels, the mean Δ NTL was -9.92 , indicating substantial darkening. However, the GMM, trained exclusively on these urban pixels, classifies only 4.5% of the full AOI as potentially damaged, because rural and suburban pixels—which constitute $> 99\%$ of the AOI area—exhibit negligible absolute NTL change. This finding underscores that VIIRS-based damage assessment must account for the urban–rural sampling imbalance.

Sentinel-2’s directional damage score shows that 42.7% of sampled pixels exhibit change in the damage direction (NDVI decrease + NBR decrease + NDBI increase), with 43.3% of those classified as high-magnitude damage by the GMM. This yields $p_2 = 0.433$, substantially lower than the omnidirectional $|PC1|$ approach (0.579 in our baseline), confirming that the direction filter eliminates false positives from vegetation growth.

SAR backscatter shows minimal change after histogram matching (mean $\Delta\sigma^0 = +0.015$ dB, down from -1.61 dB raw), with only 0.6% of pixels classified as damaged. This is consistent with the physical expectation that C-band SAR amplitude is insensitive to the structural transition from intact buildings to rubble, as both present hard, angular scattering surfaces.

5.2. Cross-Sensor Correlation Analysis

Table 3 presents the Pearson correlation matrix of damage probabilities across the four sensors. All cross-sensor correlations satisfy $|\rho| < 0.11$, with a maximum of $\rho = 0.107$ between VIIRS and SAR. This near-complete orthogonality empirically validates our central premise: each sensor captures a fundamentally different damage dimension, and single-sensor assessments are inherently incomplete.

Table 3: Cross-sensor Pearson correlation matrix of damage probabilities.

	VIIRS	S2	SAR	InSAR
VIIRS	1.000	−0.011	−0.005	0.001
S2		1.000	0.017	−0.019
SAR			1.000	0.063
InSAR				1.000

5.3. Fusion Results

The Bayesian fusion posterior (Figure 3) yields a mean damage probability of 0.176 ± 0.104 (SD of posterior means across pixels), with a mean posterior standard deviation of 0.032 and a mean 95% HDI width of 0.124. The posterior mean lies between the four single-sensor estimates, exhibiting the shrinkage-toward-prior behavior characteristic of hierarchical Bayesian models when sensors disagree.

Figure 3(b–c) reveals the spatial distribution of uncertainty. Posterior standard deviation is highest in areas where one or more sensors have missing data (e.g., InSAR coherence is only available for 6,785 of 15,000 pixels due to persistent low-coherence masking over water bodies). This demonstrates a key advantage of the Bayesian framework: uncertainty is explicitly quantified and spatially resolved, providing actionable information for humanitarian analysts who need to know not just *where* damage is estimated, but *how confident* the estimate is.

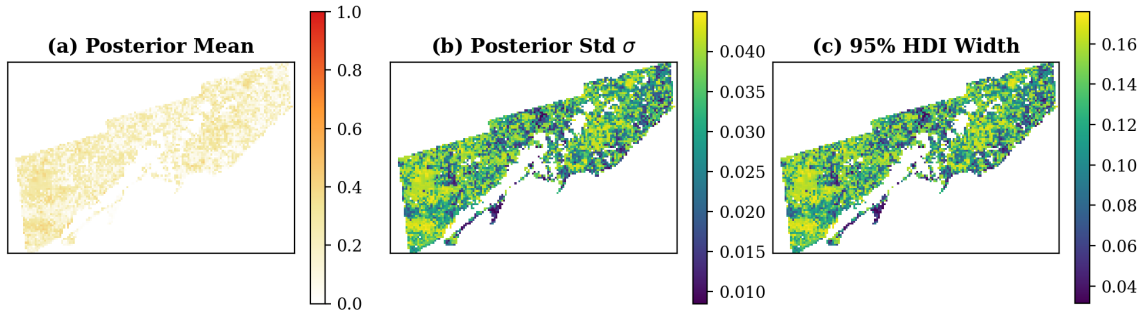


Figure 3: Bayesian fusion results: (a) posterior mean, (b) posterior standard deviation, (c) 95% HDI width.

5.4. Method Comparison

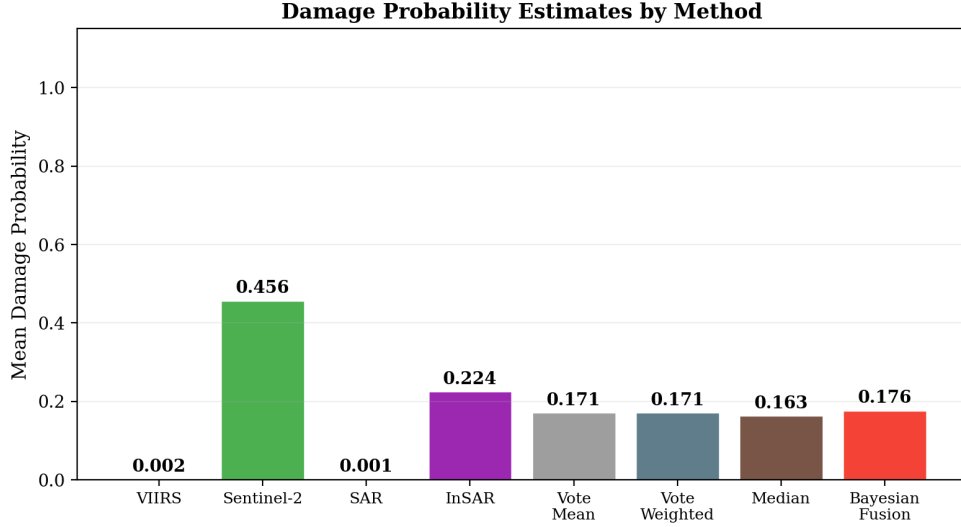


Figure 4: Mean damage probability across eight estimation methods.

Figure 4 compares the mean damage probability across all methods. The Bayesian fusion (0.176) is close to the simple mean (0.170) and precision-weighted mean (0.164), because the sensor precision parameters κ_k are similar after regularization. The Dempster-Shafer belief is substantially lower ($\text{Bel}_D = 0.014$, not shown in the bar chart), reflecting D-S’s strong penalty on conflicting evidence: when four “highly certain” sensors (low σ) provide contradictory assessments, the D-S combination rule attributes most mass to the conflict term K , driving the belief toward zero. This behavior, while mathematically correct under D-S axioms, is arguably too conservative for multi-sensor damage assessment where sensor disagreement is expected and informative rather than pathological.

The synthetic validation experiment (Table 4) confirms that the Bayesian fusion and simple voting achieve comparable AUC across five injection scenarios (0.84–0.89), with the best single-sensor AUC ranging from 0.85 to 0.89 depending on which sensor’s signal is dominant in each scenario.

Table 4: Synthetic validation results (ROC AUC) across five damage injection scenarios.

Scenario	Best Single	Vote Mean	D-S Bel	Bayesian
A: VIIRS-dominant	0.893	0.881	0.877	0.881
B: S2-dominant	0.865	0.860	0.853	0.860
C: SAR-dominant	0.875	0.857	0.858	0.857
D: All-strong	0.870	0.867	0.865	0.867
E: Mixed	0.851	0.838	0.838	0.838

5.5. Improvement Tracking

Figure 5 tracks the evolution of sensor outputs and fusion results across our three experimental versions. Version 2 (baseline) suffered from two critical flaws: (a) Sentinel-2 used winter-to-spring image pairs, introducing a $+0.14$ seasonal NDVI artifact, and (b) SAR backscatter was compared without radiometric normalization, producing a spurious -1.61 dB offset that caused $p_3 \approx 0.99$. Version 3 corrected both issues through same-season S2 pairing and histogram matching for SAR, reducing p_3 from 0.994 to 0.006. Version 4 added urban-focused VIIRS fitting (reducing dilution by dark rural pixels) and directional S2 change detection (eliminating false positives from vegetation growth), bringing p_2 from 0.579 to 0.433.

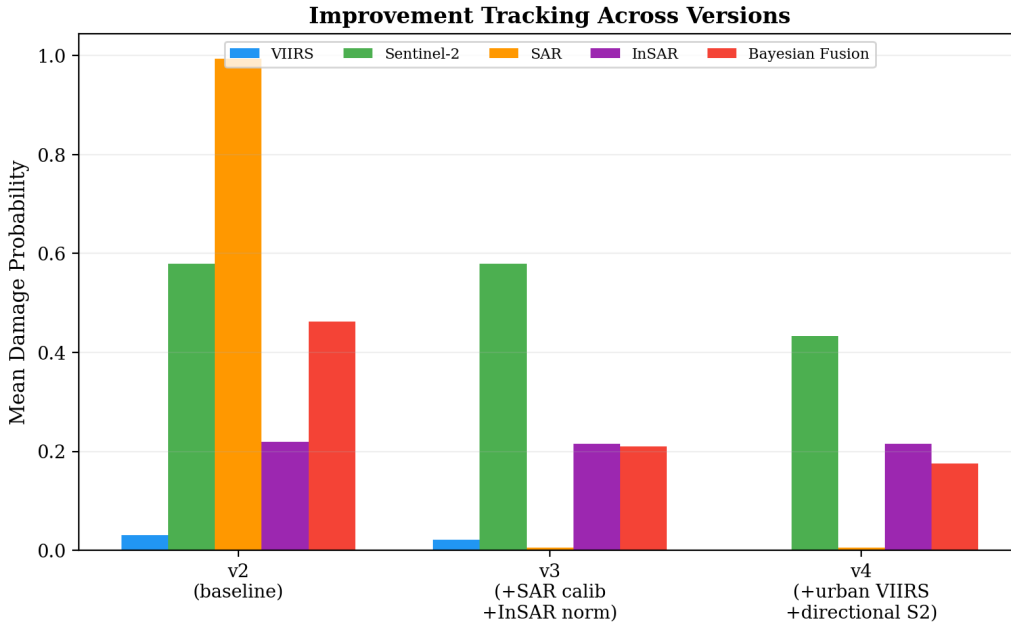


Figure 5: Improvement tracking across experimental versions. v2: baseline (uncorrected). v3: +SAR calibration, +InSAR normalization. v4: +urban VIIRS focus, +directional S2.

6. Discussion

6.1. Why Are the Sensors Near-Orthogonal?

The near-zero cross-sensor correlations ($|\rho| < 0.11$) are not a failure of our methodology but a physical necessity given the measurement principles of each instrument. VIIRS measures emitted visible light at night; Sentinel-2 measures reflected solar radiation across 13 spectral bands; Sentinel-1 measures active microwave backscatter; and InSAR measures phase coherence between repeat-pass acquisitions. These four physical quantities are coupled to

“damage” through entirely different causal pathways: electricity infrastructure collapse, surface material change, geometric scattering change, and surface displacement, respectively. The fact that they are empirically uncorrelated in Mariupol confirms that each sensor provides non-redundant information, and that single-sensor damage maps—regardless of their spatial resolution or algorithmic sophistication—are measuring only one facet of a multidimensional phenomenon.

6.2. Limitations and Future Work

Several limitations of the current study warrant discussion. First, the InSAR analysis uses only two interferometric pairs with different temporal baselines; the τ -normalization provides a physics-based correction, but a time-series approach using multiple interferograms (e.g., SBAS) would provide more robust coherence estimates. Second, the SAR histogram matching assumes that the majority of the scene is unchanged—a reasonable but unverified assumption for conflict zones. Third, the lack of authoritative building-level ground truth prevents pixel-level accuracy assessment; our synthetic validation provides a controlled benchmark, but real-world validation against UNOSAT or similar damage assessments would strengthen the findings. Fourth, the current sensor precision parameters κ_k are estimated from bootstrap resampling, which under-estimates prediction uncertainty; future work should explore calibration via Platt scaling or conformal prediction.

7. Conclusion

We presented a Bayesian hierarchical framework for fusing four heterogeneous satellite sensors—VIIRS nighttime light, Sentinel-2 multispectral, Sentinel-1 SAR, and InSAR coherence—into a unified per-pixel conflict damage probability with full uncertainty quantification. Applied to Mariupol, Ukraine, our framework produces a fused damage probability of 0.176 ± 0.032 with a 95% HDI of $[0.059, 0.297]$, and reveals that the four sensors are empirically near-orthogonal ($|\rho| < 0.11$ for all pairs). This orthogonality—far from being a limitation—is the central motivation for multi-sensor fusion: each sensor captures a dimension of damage that the others miss entirely. The Bayesian framework naturally handles sensor disagreement by increasing posterior variance, providing honest uncertainty estimates that are critical for operational humanitarian assessment in data-scarce environments. Our systematic comparison with Dempster-Shafer evidence theory reveals that D-S’s conflict penalty mechanism, while principled, is overly conservative for settings where sensor heterogeneity is both expected and informative. The improvement tracking across experimental versions (v2→v4) demonstrates that careful sensor-specific preprocessing—same-season optical pairing, radiometric SAR normalization, urban-focused NTL analysis, and directional change detection—is essential for credible multi-sensor damage assessment.